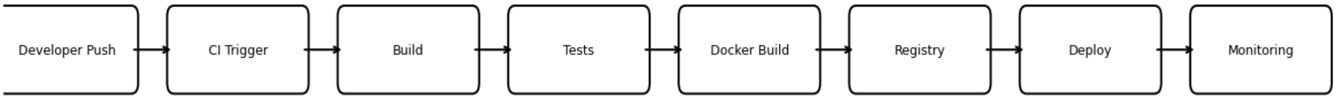
DigiCart CI/CD Architecture & Docker Deployment

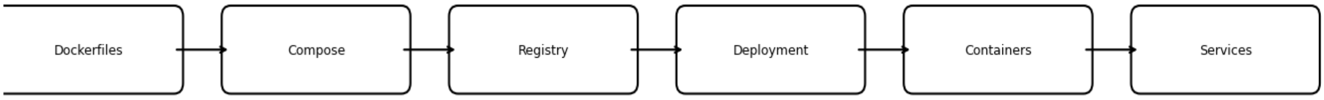
This document presents an improved CI/CD workflow and Docker deployment architecture for the DigiCart microservices platform. The diagrams illustrate the build, test, containerization, deployment, and monitoring stages used in the DevOps lifecycle.

1. High-Level CI/CD Pipeline



Stage	Description
Developer Push	Code changes pushed to repository
CI Trigger	Pipeline automatically starts
Build	Compile Java microservices
Tests	Run unit and integration tests
Docker Build	Create Docker container images
Registry	Push images to Docker registry
Deploy	Deploy using Docker Compose/Kubernetes
Monitoring	Health checks and observability

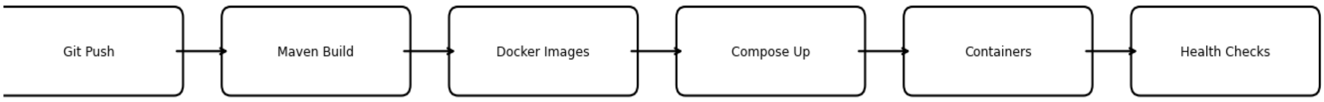
2. Docker Deployment Flow



Stage	Description
Dockerfiles	Service-specific container definitions
Compose	Docker Compose orchestration

Registry	Push images to Docker registry
Deployment	Deployment to Linux/WSL environment
Containers	Running containerized services
Services	MySQL, Kafka, Zipkin, APIs

3. WSL Docker Deployment



Stage	Description
Git Push	Developer commits source code
Maven Build	Build Spring Boot services
Docker Images	Generate deployable images
Compose Up	Start containers with compose
Containers	Running containerized services
Health Checks	Validate service availability

4. Production Release Flow

Stage	Description
Build	Compile Java microservices
Unit Tests	Execute automated tests
Integration	Validate inter-service communication

Push Images	Upload images to registry
Deploy	Deploy using Docker Compose/Kubernetes
Smoke Tests	Basic deployment verification
Release	Production-ready deployment

Recommended Deployment Commands

Build & Deploy	`docker compose up -d --build`
View Logs	`docker compose logs -f`
Stop Services	`docker compose down`
Rebuild Service	`docker compose build service-name`